# AP-S/URSI — Technical Workshop HD-7
# AI in Antenna Design and Analysis: From Pretrained Models to System-Level Integration

*Overview*

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# Agenda

## Why AI for antenna design, analysis, and optimization

**Challenges:**

- Electromagnetic analysis is very time consuming, especially when repeated across a design space
- Exploring large parametric design spaces and selecting topologies is slow and expensive
- Measuring full 3D radiation patterns is costly; often only sparse 2D cuts are available
- Optimizing antennas with many parameters is computationally intractable
- Authoring designs and connecting components to system-level simulation is fragmented and manual
- Antenna designers might not be familiar with AI

# Agenda

## Solutions addressed in this workshop

**Solutions and benefits:**

1. Pretrained AI surrogate models and AutoML
   - Rapid topology selection and design-space exploration — faster than EM techniques
2. AI-based pattern reconstruction
   - Reconstruct full 3D patterns from sparse 2D cuts using a U-Net generative model
3. AI-powered optimization
   - Surrogate-assisted and genetic-algorithm search for many-parameter designs
4. MCP-aided LLM authoring and skills workflow
   - Natural-language design intent to parametric geometry, simulation, and iteration
5. System-level integration
   - Insert antenna, RF PCB, and pattern artifacts into end-to-end RF system simulations

# Agenda

## Pretrained AI models & AI-based pattern reconstruction

**Part 1: Pretrained AI Antennas for Design Space Exploration and AutoML**

*35 minutes • Lecture + Demo + Online Exercise*

- Pretrained surrogate model library for common antenna families
- Rapid topology selection and design-space exploration using learned priors
- AutoML workflow for training custom predictors for new geometries and requirements
- Dataset generation from parametric EM simulations
- Model selection, validation, error analysis, and model-usage boundaries

**Part 2: Pattern Reconstruction with AI**

*25 minutes • Lecture + Demo + Online Exercise*

- Reconstructing full 3D patterns from sparse azimuth and elevation cuts
- U-Net-based generative model architecture and training approach
- Accuracy and speed comparison against traditional reconstruction methods
- Validation across diverse antenna types; current limitations and emerging challenges

# Agenda

## Optimization, MCP-aided LLM authoring & skills workflow

**Part 3: Optimization**

*35 minutes • Lecture + Demo/Online Exercise*

- Optimization across different antenna families; evolved pixelated topologies
- Surrogate-assisted search for rapid design-space exploration
- Customized genetic algorithm refinement for miniaturization and performance constraints
- Trade-offs among speed, accuracy, manufacturability, and EM fidelity
- Python and MATLAB co-design connecting AI, optimization, and EM simulation

**Break — 10 minutes**

**Part 3: MCP-Aided LLM-Based Antenna Authoring and Skills Workflow**

*35 minutes •*

- LLM client workflow for antenna and RF PCB design authoring
- Model Context Protocol connection between the LLM interface and engineering tools
- LLM skills for producing and modifying structured PCB geometries
- Skills for RF metrics, MATLAB code optimization, pattern analysis, and design checking
- Natural-language intent to parametric geometry, simulation, metric extraction, and iteration
- Governance: reproducibility, validation, model boundaries, and human review

# Agenda

## System integration — from component design to system simulation

**Part 4: System Integration**

*35 minutes*

- Inserting antenna, RF PCB, and pattern artifacts into system-level simulations
- Linking EM results to RF chains, waveforms, and link-budget models
- RF transmitter and receiver chain examples using antenna artifacts
- End-to-end performance evaluation for design decisions
- How LLM/MCP workflows shorten the path from component design to system simulation